SCR2043 OPERATING SYSTEMS

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Section	: Section 2	

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. **Download necessary source codes:**

   Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. **Compile the source files:**

   Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. **Execute the dummy processes:**

   Run all the dummy processes

```
./mainprocess &
```

   Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

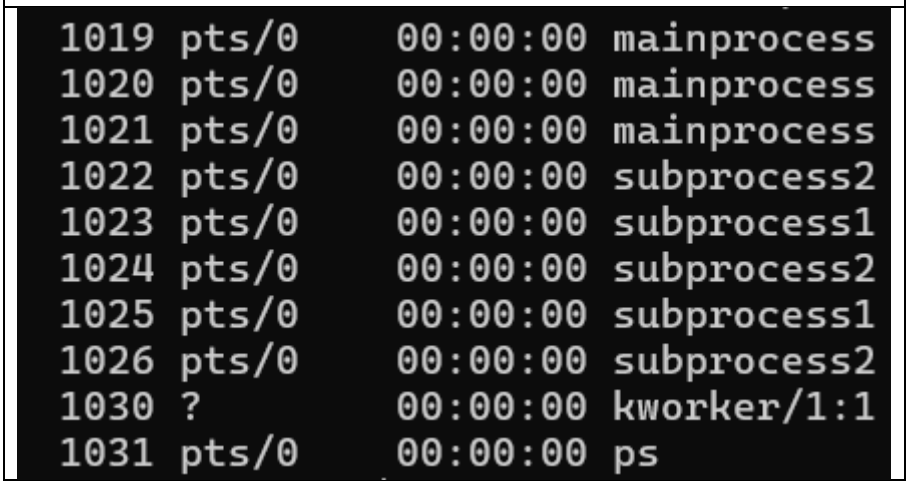
Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

Question 1

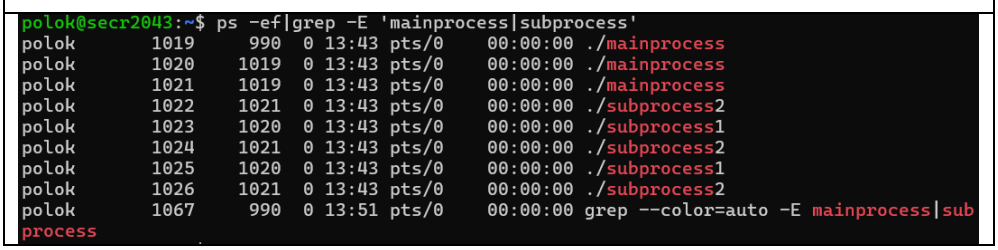
Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
`ps -e`
Output


```
1019 pts/0    00:00:00 mainprocess
1020 pts/0    00:00:00 mainprocess
1021 pts/0    00:00:00 mainprocess
1022 pts/0    00:00:00 subprocess2
1023 pts/0    00:00:00 subprocess1
1024 pts/0    00:00:00 subprocess2
1025 pts/0    00:00:00 subprocess1
1026 pts/0    00:00:00 subprocess2
1030 ?        00:00:00 kworker/1:1
1031 pts/0    00:00:00 ps
```

Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command
`ps -ef |grep -E 'mainprocess|subprocess'`
Output


```
polok@secr2043:~$ ps -ef|grep -E 'mainprocess|subprocess'
polok       1019     990  0 13:43 pts/0    00:00:00 ./mainprocess
polok       1020    1019  0 13:43 pts/0    00:00:00 ./mainprocess
polok       1021    1019  0 13:43 pts/0    00:00:00 ./mainprocess
polok       1022    1021  0 13:43 pts/0    00:00:00 ./subprocess2
polok       1023    1020  0 13:43 pts/0    00:00:00 ./subprocess1
polok       1024    1021  0 13:43 pts/0    00:00:00 ./subprocess2
polok       1025    1020  0 13:43 pts/0    00:00:00 ./subprocess1
polok       1026    1021  0 13:43 pts/0    00:00:00 ./subprocess2
polok       1067     990  0 13:51 pts/0    00:00:00 grep --color=auto -E mainprocess|sub
process
```

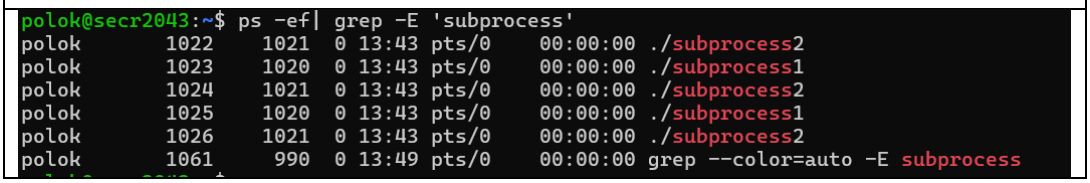
Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command

```
ps -ef| grep -E 'subprocess'
```

Output



Question 4

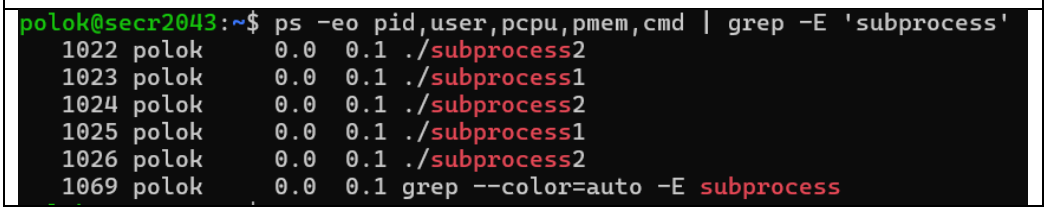
Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command

```
ps -eo pid,user,pcpu,pmem,cmd | grep -E 'subprocess'
```

Output



Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

Command

```
ps -eo pid,user,pcpu,pmem,cmd --sort=pmem | grep -E 'subprocess'
```

Output

```
polok@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd --sort=pmem | grep -E 'subprocess'
   1022 polok     0.0  0.1 ./subprocess2
   1023 polok     0.0  0.1 ./subprocess1
   1026 polok     0.0  0.1 ./subprocess2
   1024 polok     0.0  0.1 ./subprocess2
   1025 polok     0.0  0.1 ./subprocess1
   1085 polok     0.0  0.1 grep --color=auto -E subprocess
```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command

```
ps --forest -C mainprocess -C subprocess1 -C subprocess2
```

Output

```
polok@secr2043:~$ ps --forest -C mainprocess -C subprocess1 -C subprocess2
    PID TTY          TIME CMD
   1019 pts/0    00:00:00 mainprocess
   1020 pts/0    00:00:00  \_ mainprocess
   1023 pts/0    00:00:00  |   \_ subprocess1
   1025 pts/0    00:00:00  |   \_ subprocess1
   1021 pts/0    00:00:00  \_ mainprocess
   1022 pts/0    00:00:00      \_ subprocess2
   1024 pts/0    00:00:00      \_ subprocess2
   1026 pts/0    00:00:00      \_ subprocess2
```

Question 7

Use `pstree` command with option that show the number of threads to each process.

Command

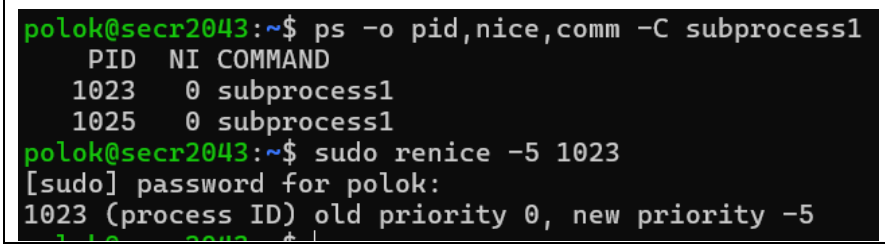
```
pstree -c -s 1019
```

Output

```
polok@secr2043:~$ pstree -c -s 1019
systemd---sshd---sshd---sshd---bash---mainprocess-+-mainprocess-+-subprocess1
                                                   |             `-subprocess1
                                                   `-mainprocess-+-subprocess2
                                                                 |-subprocess2
                                                                 `-subprocess2
```

Question 8

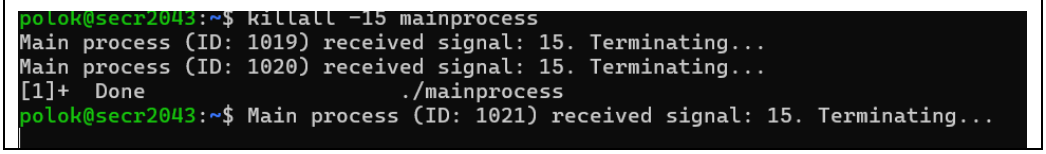
Use `renice` command to change priority level of one of process “subprocess1”.

Command
`sudo renice -5 1023`
Output


```
polok@secr2043:~$ ps -o pid,nice,comm -C subprocess1
    PID  NI COMMAND
   1023   0 subprocess1
   1025   0 subprocess1
polok@secr2043:~$ sudo renice -5 1023
[sudo] password for polok:
1023 (process ID) old priority 0, new priority -5
```

Question 9

Terminate all running processes with the name “mainprocess”.

Command
`killall -15 mainprocess`
Output


```
polok@secr2043:~$ killall -15 mainprocess
Main process (ID: 1019) received signal: 15. Terminating...
Main process (ID: 1020) received signal: 15. Terminating...
[1]+  Done                    ./mainprocess
polok@secr2043:~$ Main process (ID: 1021) received signal: 15. Terminating...
```

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

void child_process() {
	printf("Child process with PID: %d\n", getpid());
	int sleep_time = rand() % 5 + 1;
	printf("Child process sleeping for %d seconds\n", sleep_time);
	sleep(sleep_time);
	printf("Child process exiting\n");
}

int main() {
	printf("Parent process with PID: %d\n", getpid());

	// Fork a child process
	pid_t pid = fork();

	if (pid == 0) {
		// This is the child process
		child_process();
		exit(0);
	} else if (pid > 0) {
		// This is the parent process
		printf("Parent process waiting for child process to finish\n");
		// Wait for the child process to finish
		wait(NULL);
		printf("Parent process exiting\n");
	} else {
	// Error occurred while forking
		perror("fork");
		return 1;
	}
	return 0;
}
```

```
 GNU nano 7.2                                  example.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

void child_process() {
        printf("Child process with PID: %d\n", getpid());
        int sleep_time = rand() % 5 + 1;
        printf("Child process sleeping for %d seconds\n", sleep_time);
        sleep(sleep_time);
        printf("Child process exiting\n");
}

int main() {
        printf("Parent process with PID: %d\n", getpid());

        // Fork a child process
        pid_t pid = fork();

        if (pid == 0) {
                // This is the child process
                child_process();
                exit(0);
        } else if (pid > 0) {
                // This is the parent process
                printf("Parent process waiting for child process to finish\n");
                // Wait for the child process to finish
                wait(NULL);
                printf("Parent process exiting\n");
        } else {
        // Error occurred while forking
                perror("fork");
                return 1;
        }
        return 0;
}
```

Output:

```
polok@secr2043:~$ nano example.c
polok@secr2043:~$ gcc example.c -o example
polok@secr2043:~$ ./example
Parent process with PID: 1116
Parent process waiting for child process to finish
Child process with PID: 1117
Child process sleeping for 4 seconds
Child process exiting
Parent process exiting
```